Money and Crypto Portal Project Report

Project overview
My project is a responsive web application for exploring EUR/USD exchange rates and cryptocurrency prices. It includes monthly charts for EUR/USD, Bitcoin, Dogecoin and Ethereum, chart controls, a portfolio calculator, a live converter, and chart export and sharing controls. The application is built with HTML, CSS and JavaScript, with Frappe Charts for the visualizations.

Data and implementation
The application uses three independent data services. Frankfurter supplies daily EUR/USD rates, which the application groups into monthly averages. CryptoCompare supplies daily cryptocurrency prices in USD; the application takes each month's closing price and converts it to EUR using that month's Frankfurter rate. CoinGecko supplies current EUR and USD prices for the live converter and the portfolio calculator. Combining the CryptoCompare and Frankfurter data creates monthly cryptocurrency prices in EUR for the charts.
The charts support six months, ten years or twenty years of data. Users can select a currency color, drag currencies onto charts, add charts, and click chart points to see monthly values. A chart can show multiple currencies as separate lines. New charts are placed next to the selected chart and begin with instructions to add a currency. The portfolio calculator estimates the current value of a past crypto purchase and gives a rule-based suggestion based on the change in value. Users can download the selected chart as an SVG. The sharing controls download the chart and open a social media composer; the user attaches the downloaded image before posting.

User instructions
Choose a time period with the buttons at the top. To create or compare chart lines, drag one or more currencies from the sidebar onto a chart. Select a currency in the color control and choose a color to update that currency's line. Click a chart to make it the selected chart; use Add Chart Next to Current to insert an empty chart beside it, or Remove to delete an added chart. Click a data point to see its monthly values. Use the download or sharing controls for the selected chart. To use the converter, enter an amount, choose a cryptocurrency and select Convert Now. To estimate a purchase, enter its date and amount, choose a cryptocurrency and select Calculate Value and Advice.

Requested points and reasons
I request 40 points. My self-assessment is based on the following rubric items and features:
Well-written project report: 3 points. This report describes the application, data sources, tools, features and user instructions, and includes the required AI declaration.
Responsive design: 4 points. The CSS includes flexible layouts and a mobile breakpoint so the controls, charts and calculator can adapt to narrow screens.
Firefox, Safari, Edge and Chrome: 3 points requested. The application uses standard HTML, CSS and JavaScript with Frappe Charts. I will check the finished project in all four required browsers before submission.
Clear directory structure: 2 points. The application separates its HTML, CSS and JavaScript into index.html, styles.css and script.js.
Project features: drag-and-drop charts, 4 points; time-period selection, 2; currency colors, 2; adding and removing charts, 2; point details, 2; three data sources, 5 points across the two API-source rubric items; combining exchange-rate and crypto data, 3; purchase calculator and recommendation, 5; SVG download, 2; social sharing preparation, 2.

Additional feature: 1 point requested for the live EUR converter, which converts a user-entered amount of BTC, DOGE or ETH using current CoinGecko prices. This brings my requested total to 40 points.

AI use declaration

I used ChatGPT (OpenAI) during the planning, coding and documentation of this project. ChatGPT helped write and revise JavaScript, not only proofread it. It helped implement the $() helper for finding page elements and bindEvents() for connecting buttons, color controls and drag-and-drop to their event handlers.

It also assisted with the startApp(), rendering charts and displaying point details, and connecting the portfolio calculator and converter to the data. I reviewed and adapted the changes/suggestions in my project.

I also used ChatGPT for suggestions on input validation, error handling and user-facing text, including the portfolio calculator messages and its rule-based buy, hold or sell advice. ChatGPT reviewed the chart behavior and helped with resizing charts when placing new charts beside the selected ones, point details for all lines, and the download and sharing updates. ChatGPT helped draft and organize this report. I reviewed the suggestions and take responsibility for the submitted code and report.

ChatGPT also helped me organize parts of the HTML and CSS. In the HTML, it suggested how to structure the page into sections for the controls, chart area, data point information, converter, portfolio calculator, and export and sharing buttons. In the CSS, it helped with arranging the controls and charts, styling buttons, chart areas, and form fields. I reviewed and adapted these suggestions for my project.

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